DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

# NITK, Surathkal

**Algorithms And Complexity (Code: CS700)**
**I Sem MTech. CSE**
**October 06, 2025**

Registration No. 252CS042

Time: 1:30 hour — *Mid Sem Exam* — Maximum Marks: 40

**Note:-** *Answer all the questions. Missing data may be suitably assumed, provided the assumption does not change the difficulty of the problem.*

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1. (a) The Recurrence $T(n) = 7T(n/2) + n^2$ describes the running time of an algorithm $A$. A Computing algorithm $A'$ has a running time $T'(n) = aT'(n/4) + n^2$. What is the largest value for '$a$' such that $A'$ is asymptotically faster than $A$ ?

   (b) Can the master method be applied to the recurrence $T(n) = 4T(n/2) + n^2 logn$ ? Why or why not ? give an asymptotic upper bound for this recurrence.

[04+06]

2. Write the recurrence for best, worst and average case complexity of quick sort algorithm. Also derive its average case analysis. [08]

3. Suppose we have an array of sorted integers that has been circularly shifted k positions to the right. For example, [35, 42, 5, 10, 20, 30] is a sorted array that has been circularly shifted $k = 2$ positions, while [27, 29, 35, 42, 5, 9] is a sorted array that has been shifted $k = 4$ positions. Now, suppose you are given a sorted array that has been shifted an unknown number of times – we do not know what $k$ is. Describe how you would implement an algorithm to find $k$ in $O(log(n))$ time. [08]

4. Design an $O(n \log n)$ algorithm to find the closest pair of points out of n points in a two dimensional space.

[06]

5. While walking on the beach one day you find a treasure trove. Inside there are n treasures with weights w1,w2..... wn and cost c1, c2,..... cn. Unfortunately you have a bag that only holds a total weight M. Fortunately there is a knife handy so that you can cut treasures if necessary.
   a) Describe a $\theta(nlogn)$ time greedy algorithm to solve this problem
   b)Prove the correctness of your algorithm. [08]

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Algorithms And Complexity (Code: CS700)

November 17, 2025

I Semester M.Tech CSE

Registration No.

Time: 60 minutes. *Quiz-2* Maximum Marks: 20

**Note:-** *Answer all the questions. Missing data may be suitably assumed, provided the assumption does not change the difficulty of the problem.*

1. Design an efficient algorithm for the matrix chain multiplication problem that outputs a fully parenthesized expression for how to multiply the matrices in the chain using minimum number of operations. [08]

2. Give a polynomial time reduction from SAT to CNF problem. [06]

3. Give a polynomial time reduction from 3-SAT to Vertex Cover problem. Prove that Vertex Cover is NP-Hard. [06]



# DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

## NITK, Surathkal

**Algorithms And Complexity (Code: CS700)**

**November 28, 2025**

I Semester M.Tech CSE

Registration No.

Time: 3Hrs. *End Semester Exam* Maximum Marks: 80

**Note:-** *Answer all the questions. Missing data may be suitably assumed, provided the assumption does not change the difficulty of the problem.*

1. In the algorithm SELECT(the $k^{th}$ smallest in a set whose worst case linear time), the input elements are divided into groups of 5. Will the algorithm work in linear time, if they are divided into groups of 7? How about groups of 3? Analyze the running time of this algorithm? [06]

2. Let $X[1 \ldots n]$ and $Y[1 \ldots n]$ be two arrays each containing $n$ numbers already in sorted order. Design and analyse an efficient divide and conquer $O(\log n)$-time algorithm to find the median of all $2n$ elements in arrays $X$ and $Y$. [08]

3. Given $n$ files of length $m_1, m_2, \ldots\ldots m_n$, the Optimal Tape Storage problem is to find which order is the best to store them on the tape, assuming that
   i) Each retrieval starts with the tape rewound.
   ii)Each retrieval takes time equal to the length of the preceding files in the tape plus the length of the retrieval file

   a) Describe a greedy algorithm for this problem.
   b) What is the running time of your algorithm?
   c) Prove your algorithm is correct. [10]

4. *Given a sequence* $k = k_1 < k_2 < \ldots\ldots < k_n$ of $n$ sorted keys, with a search probability $p_i$ for each key $k_i$ . Build the Binary search tree that has the least search cost given the access probability for each key. Design an algorithm for the above given problem.

   Show the step by step method to construct an optimal BST for keys $A, B, C$, and $D$ with search probabilities 0.1, 0.2, 0.4, and 0.3, respectively? [12]

5. Construct a maximum length common sub-sequence from the completed C table and the original sequences $X =< A, B, C, B, D, A, B >$ and $Y =< B, D, C, A, B, A >$ Show all the steps. [08]

6. Give a polynomial time reduction from CNF to 3-SAT problem. [07]

7. Assuming that Hamiltonian cycle problem is NP-Complete, prove that travelling salesman problem (TSP) is NP-Complete. [08]

8. Give a 2 - factor approximation algorithm with analysis for vertex cover problem. [10]

9. a. Solve the recurrence $T(n) = 2T(\sqrt{n}) + \log n$ and give tight bounds.
   b. Let A1, A2, A3, and A4 be four matrices of dimensions 30 × 1, 1 × 40, 40 × 10 and 10 × 25, respectively. The minimum number of scalar multiplications required to find the product $A1A2A3A4$ using the basic matrix multiplication method is .....
   c. Define NP-Completeness
   d. Consider the definition of an approximation algorithm. Explain the meaning of approximation ratio in the case of a maximisation problem.

[Each question 02]

10. Answer the following questions with either TRUE or FALSE. Explain your choice. (No credit if no explanation given.)

   (a) To prove the correctness of a greedy algorithm, we must prove that every optimal solution contains our greedy choice.

   (b) 3SAT cannot be solved in polynomial time, even if P = NP.

   (c) Repeatedly selecting a vertex of maximum degree, and deleting the incident edges, is a 2-approximation algorithm for Vertex Cover.

[01 marks each]

# DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

## NITK, Surathkal

**Algorithms And Complexity(Code: CS700)**
**I Sem M Tech. CSE**
**September 22, 2025**

Registration No.

| 2 | 5 | 2 | C | S | 0 | [illegible] | [illegible] |
|---|---|---|---|---|---|---|---|

Time: 1 hour

*Quiz - 1*

Maximum Marks: 20

**Note:-** *Answer all the questions. Missing data may be suitably assumed, provided the assumption does not change the difficulty of the problem.*

1. (a) Is $2^{n+1} = O(2^n)$ ?
   (b) Is $2^{2n} = O(2^n)$ ?

[02+02]

2. Solve the following recurrences. Assume that $T(n)$ is constant for sufficiently small $n$.
   a. $T(n) = 3T(\sqrt{n}) + logn$
   b. $T(n) = 2T(\frac{n}{2}) + \frac{n}{\log n}$

[02+03]

3. Use recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = T(\frac{n}{4}) + T(\frac{n}{2}) + n^2$. Use the substitution method to verify your answer. [04]

4. Write a divide-and-conquer algorithm that finds the maximum difference between any two elements of a given array of $n$ numbers . Prove the lower bound of comparisons in worst case. For example, on input A=[4.5, 10, -2, $\pi$, -7.115] your algorithm should return 17.115. [07]

$\frac{22}{7} = 3.14$
$-7.115 - 3.14$
$10 + 7.115.$